

Sample End Semester Question Paper - 2025

Subject: Design and Analysis of Algorithms | Code: CS502 | Semester: 5

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks: 50

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. State one use of backtracking.
2. What is a recurrence relation?
3. What is a heap?
4. What is the time complexity of merge sort?
5. State the greedy-choice property.
6. State the Master theorem briefly.
7. Name the three steps of divide and conquer.
8. What is the knapsack problem?
9. Name two algorithms for minimum spanning tree.
10. What is a substructure in dynamic programming?

PART B - Long questions (8 x 10 = 80 Marks)

1. Explain dynamic programming. Solve the 0/1 knapsack with an example.
2. Solve the longest common subsequence problem using dynamic programming.
3. Explain backtracking. Solve the N-Queens problem.
4. Compare greedy and dynamic programming approaches with examples.
5. Solve a given recurrence using the Master theorem and the recursion tree method.
6. Explain the greedy method. Solve the fractional knapsack with an example.
7. Explain Prim's and Kruskal's MST algorithms. Solve a given graph.
8. Explain Dijkstra's shortest path algorithm. Solve a given graph.